Pratirup

A glucose digital twin for type 2 diabetes care teams

प्रतिरूप (pratirūp): counterpart, likeness

Team Aflatoon · Scaler School of Technology, Bengaluru
Dhruv Sharma · Pratyush Mishra · Anushika Chauhan
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Research prototype. Not a medical device. All results are retrospective, on open research data.

What we built and what it achieves

−26%

60-min forecast error vs last value carried forward on 20 unseen patients (95% CI -6.46 to -4.24 mg/dL)

81%

of outcomes fell inside the 80% prediction interval (60 min): the stated uncertainty is honest

65%

of rises above 180 mg/dL flagged ≥ 30 min ahead, at 1.51 false alerts per patient-day

- A per-patient twin: clinical record + CGM + medication and meal logs, updated on every reading, scoring every forecast when its outcome arrives
- A clinician worklist, an explainable patient view, a forecast ledger, a what-if labelled as associational, live streaming, FHIR R4 records
- Validated twice: held-out ShanghaiT2DM patients (scored once, locked) and an external US cohort (CGMacros): 60-min MAE 16.9 vs 22.4 mg/dL in T2D
- Open: code, trained models, data pipeline and the scripts that regenerate every number on these slides

Diabetes in India is common and poorly controlled

101 M

adults with diabetes in India (ICMR-INDIAB-17, 2023)

136 M

with prediabetes (same survey)

36%

of people with diagnosed diabetes reach HbA1c < 7% (ICMR-INDIAB-13, 2022)

< 25%

consensus target for time above 180 mg/dL (Battelino 2019)

How CGM is reviewed today

CGM use is growing in India through clinic-placed and short-term sensors. Clinicians review an ambulatory glucose profile after the fact, at a visit, separately from labs, complications and the drug regimen.

The gap we address

Nothing in a typical review combines the record with the live stream, tells the care team who is likely to go high or low in the next hour, or shows how trustworthy such a forecast has been for this patient.

Forecasting already exists. What is still missing

Product	What it predicts	Who sees it	Uses clinical record?
Dexcom G7	Urgent low within 20 min	Patient	No
Medtronic Guardian	High / low up to 60 min	Patient	No
Roche Accu-Chek SmartGuide	Low in 30 min, 2-h trajectory, night low	Patient	No (CGM + logged carbs, insulin)
LibreView, Dexcom Clarity, Glooko	Retrospective reports	Clinician	No
Pratirup	**+30 / +60 min (+120 exploratory), P(>180, >250, <70 in 60 min)**	**Care team, ranked panel**	**Yes, plus own error history**

We do not claim forecasting as our novelty. What we did not find is a clinician-facing view that fuses the record, ranks a panel by imminent risk, explains each forecast and keeps a running score of its own accuracy.

Sources: vendor documentation (Dexcom, Medtronic, Roche, Glooko), read Oct 2026. Vendor claims are not independent evidence of clinical benefit.

Use case: a care team running a 14-day CGM programme

1

Wear

Adult with T2D wears a CGM for 14 days; readings sync every 15 min

2

Update

The twin updates its state on every reading and logged meal or dose

3

Triage

The worklist ranks patients: low now, predicted low, predicted > 180 or > 250

4

Act

The educator reviews the forecast and its reasons, and calls the patient if needed

5

Review

At the visit, the doctor replays any moment: what the twin expected vs what happened

Outputs

Primary: glucose at +30 and +60 min with an 80% prediction interval (+120 min exploratory). Secondary: calibrated probability of crossing 180 or 250, or falling below 70 mg/dL within 60 min (≥ 2 consecutive readings).

Boundaries

Clinicians make every decision. No dosing, diet or treatment advice. Whether triage improves outcomes is a hypothesis that needs a prospective study.

What makes it a twin: a state that updates on every reading

Clinical context

observed

Age, BMI, duration, HbA1c, C-peptide, eGFR, complications, drug classes (FHIR R4)

Glycaemic state

estimated

Kalman level, rate of change and their uncertainty; grows during sensor gaps

Event context

observed

Time since meals and fast insulin, insulin on board, oral agents, pump rate

Personal profile

estimated

Time in range so far; what this person's glucose usually does at this hour

Model trust

estimated

Running mean of this patient's own resolved forecast errors

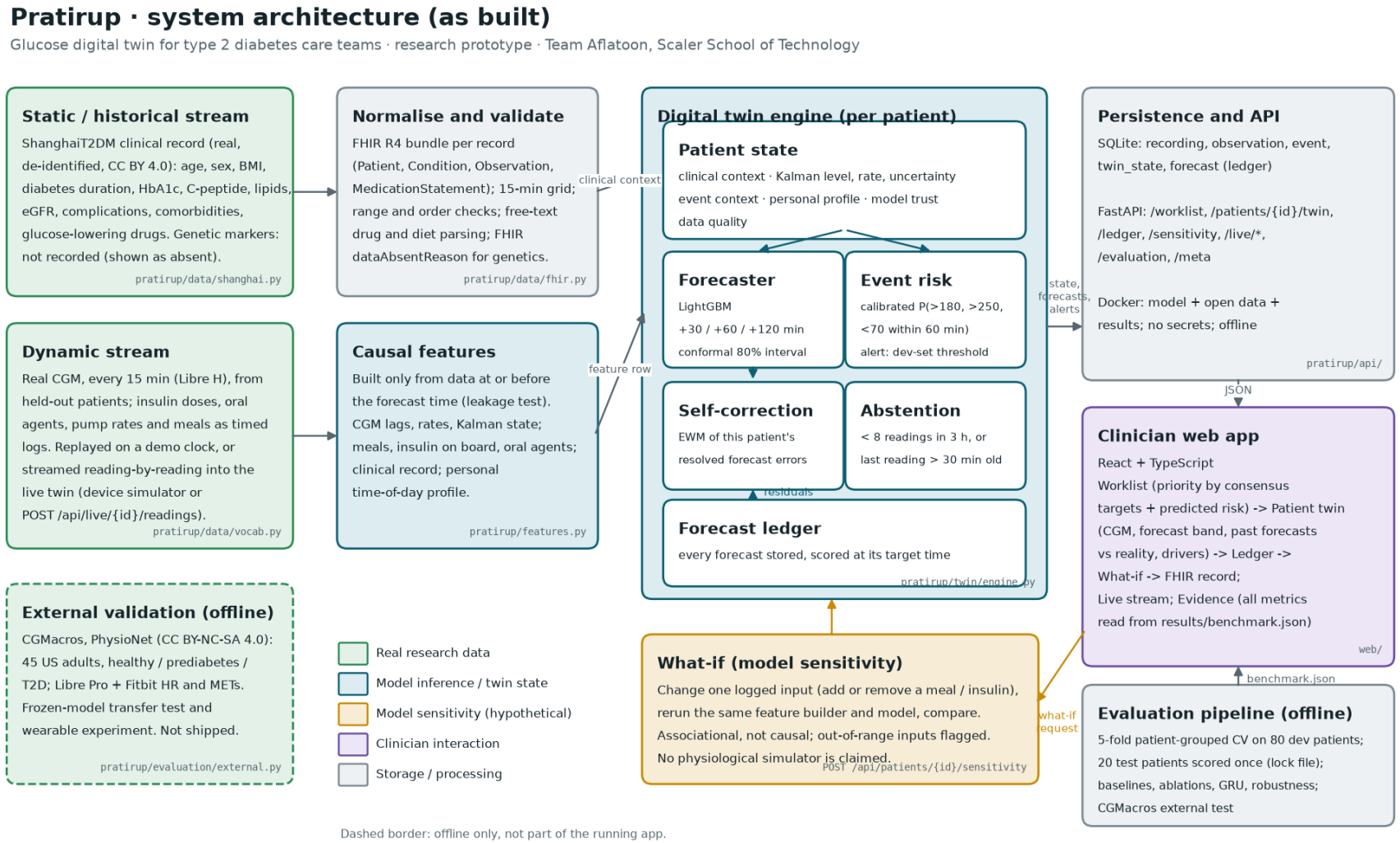
Data quality

observed

Readings in the last 3 h, age of last reading: abstain instead of guessing

Update loop on each reading: validate → Kalman update → resolve forecasts whose time has come → rebuild features from the past only → forecast + interval + risk → write to ledger → re-rank worklist. The streaming twin and the evaluated replay give identical outputs (parity test).

Architecture (as built)



Data: the two streams belong to the same patients

ShanghaiT2DM: train, test and demo

100 adults with T2D, 109 recordings of 3–14 days. Libre H CGM every 15 min (112,462 readings). Linked clinical record (33 fields) plus timed insulin, oral-agent, pump and diet logs. Zhao et al., Scientific Data 2023. CC BY 4.0.

CGMacros: external validation only

45 US adults (15 healthy, 16 prediabetes, 14 T2D), Libre Pro, Fitbit heart rate and activity, meal times. Free-living. Das et al., Scientific Data 2025 (PhysioNet). CC BY-NC-SA 4.0, not redistributed.

Rejected

- Synthea + an unrelated CGM dataset: records and readings would belong to different people
- UCI Diabetes #34: type 1 diabetes, finger-prick readings, around 1991
- AI-READI: no redistribution right; fully synthetic physiology: tests only our own assumptions

Honest gaps

- No open Indian CGM dataset exists: Chinese hospital cohort, partly in-hospital
- No genetic markers anywhere: shown as 'not recorded', never invented
- Diet is food weight, not nutrients (carbohydrate-dense-weight heuristic)

Models and an evaluation protocol fixed before testing

Compared models

Last value carried forward (persistence)
Kalman damped trend
Linear autoregression (ridge)
GRU sequence network (tuned on development data)
LightGBM ladder: CGM → + logs → + clinical record → + personal history = Pratirup
Pratirup + online self-correction
Average of LightGBM and GRU (reported, not deployed)

Served: LightGBM. It ties the GRU and gives exact per-feature attributions, conformal intervals and fast CPU inference.

1

Patient-level split

80 development patients (5 folds, stratified by insulin) and 20 test patients

2

Every choice on development data

Kalman, hyperparameters, GRU, conformal adjustment, calibration, alert thresholds

3

No leakage

Early stopping on held-out dev patients; a test proves features use only the past

4

Test scored once

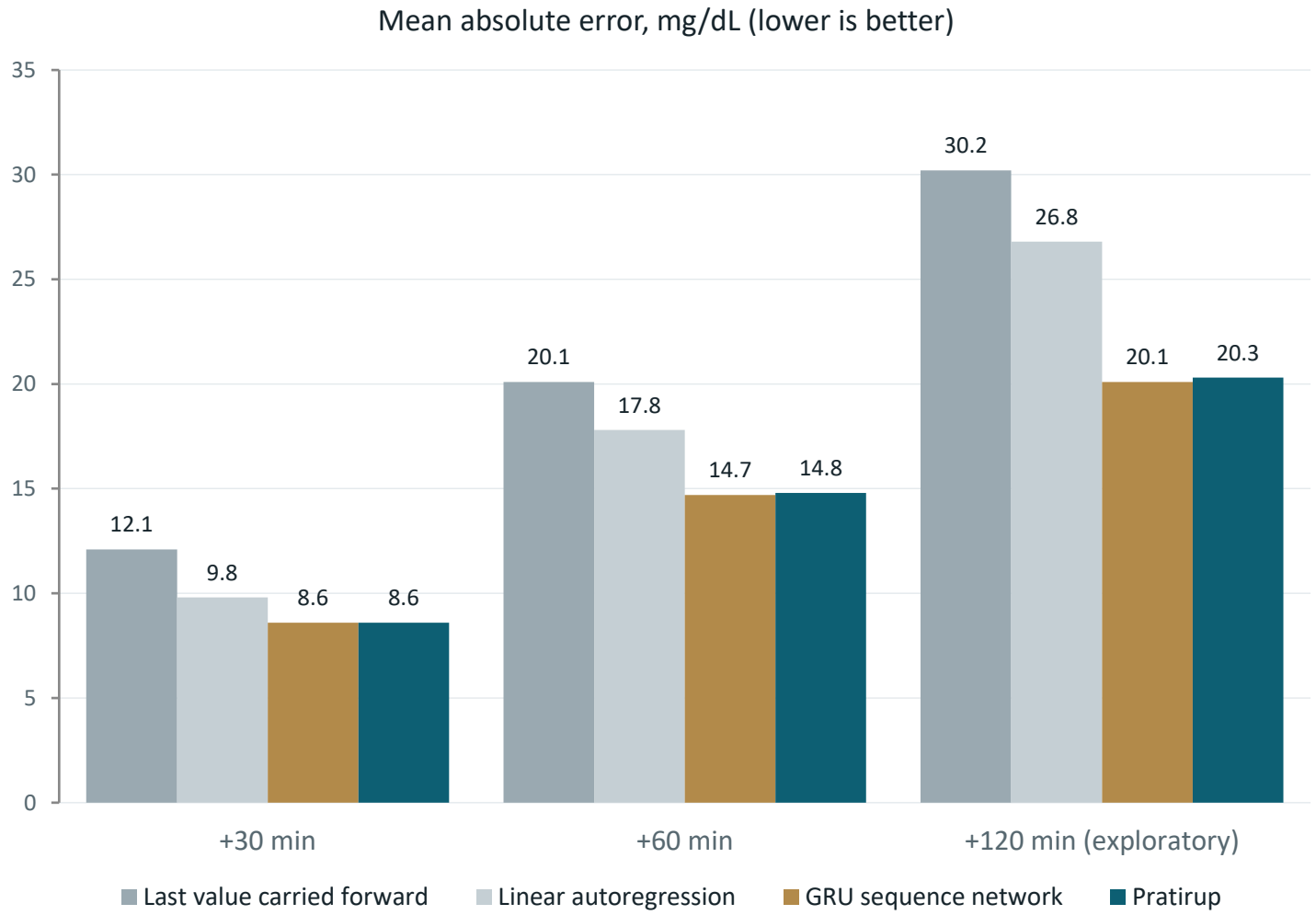
Locked file records when; reruns need a logged reason

5

Uncertainty of results

Patient-level bootstrap 95% CIs for every comparison

Forecast accuracy on 20 unseen patients



vs last value carried forward

+30 min: -3.44 mg/dL (95% CI -4.10 to -2.81)

+60 min: -5.32 mg/dL (95% CI -6.46 to -4.24)

+120 min: -9.84 mg/dL (95% CI -11.85 to -8.13)

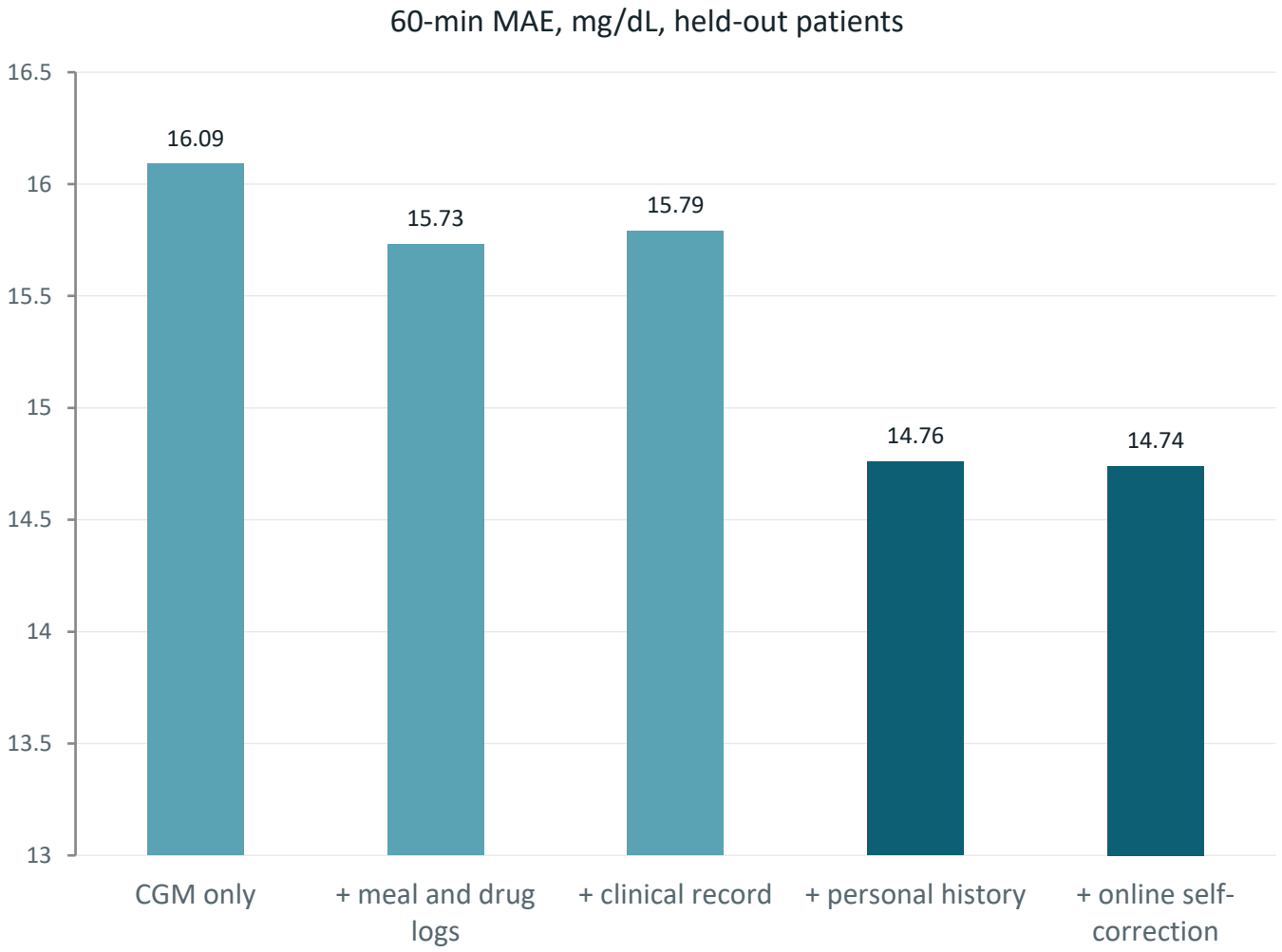
Clinical safety

Parkes type-2 error grid, zones A+B: 100.0% / 100.0% / 99.4% at 30 / 60 / 120 min

GRU vs Pratirup

No measurable difference at 60 min: 0.04 mg/dL (95% CI -0.14 to 0.20). We serve the model we can explain.

What each data stream adds, including what did not help



Helped most: the patient's own history

What this person's glucose usually does at this time of day, learned from their previous days. That is the twin's personal state at work.

Did not help measurably

Clinical-record fields beyond CGM and logs; online self-correction (-0.02 mg/dL, CI crosses 0); Fitbit heart rate and activity in CGMacros (-0.02 mg/dL, CI crosses 0). We report this rather than hide it.

Honest uncertainty: the 80% band holds about 80% of the time

81.3%

of outcomes inside the 80% interval at +30 min (mean width 27.3 mg/dL)

80.9%

of outcomes inside the 80% interval at +60 min (mean width 45.6 mg/dL)

83.5%

of outcomes inside the 80% interval at +120 min (mean width 67.2 mg/dL)

How the interval is built

Quantile models give a raw band; conformal calibration on out-of-fold predictions from development patients widens it just enough to reach 80%. The test patients were never used.

Why it matters clinically

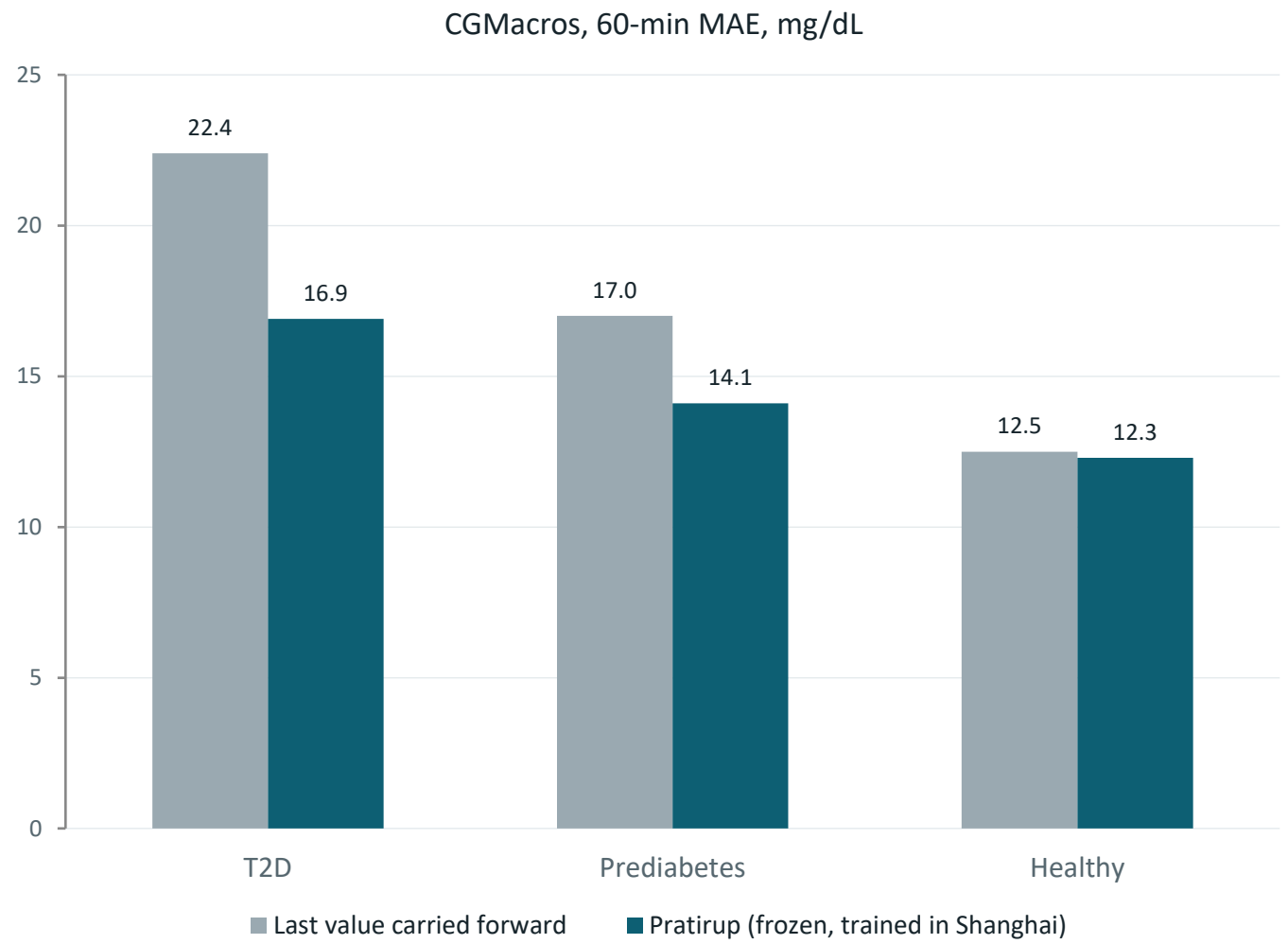
A care team can trust the band, not just the line. It held on CGMacros too: 78% in T2D, 81% in prediabetes. Under heavy sensor noise coverage falls (74% at ±10 mg/dL), a known limitation.

Early warning: more episodes caught, earlier

Event within 60 min	Scorer	Episodes	Sensitivity	≥ 30 min warning	PPV	False alerts / patient-day	AUPRC (prevalence)
Rise above 180 mg/dL	Pratirup	288	95.5%	64.9%	46.9%	1.51	0.55 (0.049)
	Rate-of-change rule		89.2%	47.6%	40.8%	1.81	0.40 (0.049)
Rise above 250 mg/dL	Pratirup	54	98.1%	81.5%	15.6%	1.39	0.55 (0.008)
	Rate-of-change rule		98.1%	77.8%	12.6%	1.79	0.32 (0.008)
Fall below 70 mg/dL	Pratirup	76	94.7%	64.5%	10.2%	3.08	0.34 (0.011)
	Rate-of-change rule		73.7%	46.1%	9.5%	2.58	0.12 (0.011)

Thresholds were set on development patients for ≤ 2 false alerts per patient-day, with a 60-min cooldown. On test patients the low-glucose alert overshot (3.08 per day), and the three alert types together produce 7.9 alerts per patient-day: too many for routine use. Next step: one merged alert stream with a shared cooldown, and budgets set with clinicians.

It transfers: external validation on a US cohort



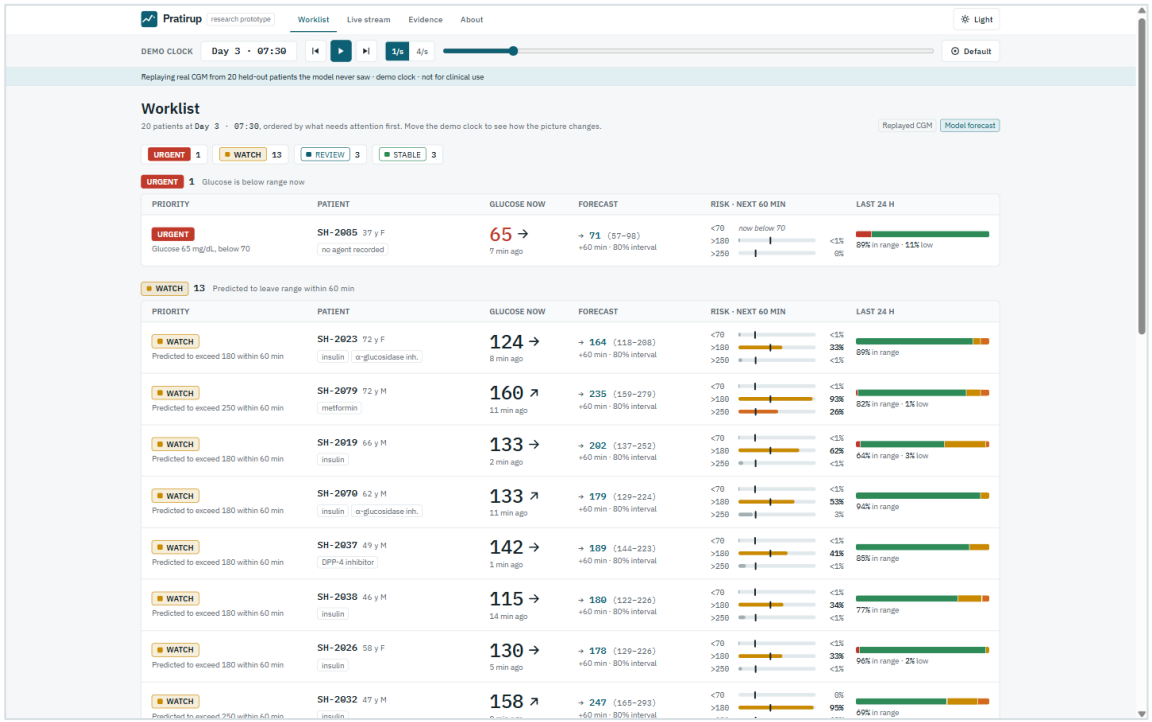
Different country, sensor and setting

Free-living US adults, Libre Pro, no refitting. Interval coverage stays at 78% in T2D. Rise-above-180 alerts in T2D: 96.7% sensitivity, 53% PPV, 1.38 false alerts per day.

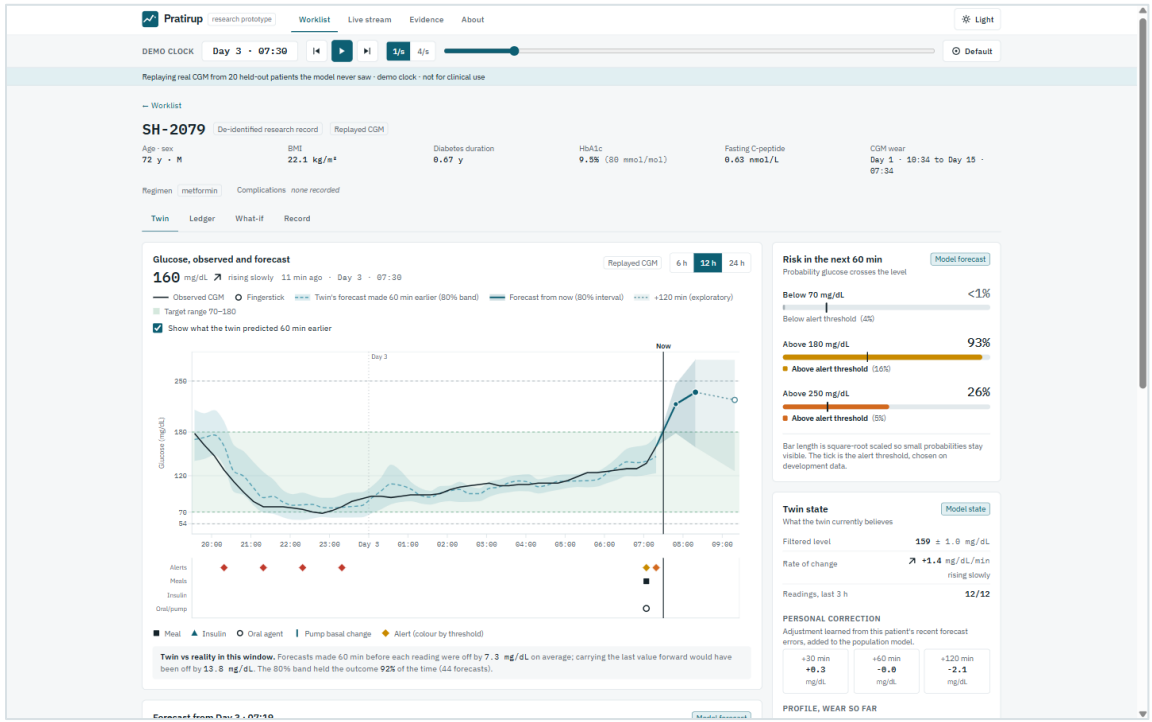
Where it does not help

In healthy participants the gain over last value is close to zero: expected for a model trained on T2D. Medications and most labs are missing in CGMacros and were passed as missing, never as zero.

The product: what a care team sees



Worklist: panel ranked by consensus targets and predicted risk



Patient twin: forecast band, past forecasts vs reality, drivers

Research record | Replayed CGM | Model forecast | Sensitivity · not causal | LIVE stream

Also: forecast ledger · what-if · FHIR R4 record · live streaming · evidence page

Limitations we want the jury to know

Population

Chinese hospital cohort, 2019–2021, partly in-hospital with regular meal times. Not Indian. External check covers the US, not India.

Size

100 patients, 3–14 days each; 20 test patients. Hence bootstrap CIs on every claim.

Alert load

7.9 alerts per patient-day across three types; low-glucose alert above its false-alert budget.

Sensor noise

With ±10 mg/dL random noise added, the 30-min forecast is worse than last value (19.6 vs 15.2). Libre output is pre-smoothed; noisier devices need noise-augmented training.

Causality

The what-if shows model sensitivity, not the effect of eating or dosing.

Evidence level

Retrospective only. No prospective, real-time or outcome study.

Path to a real product

Who would use and buy it (to validate)

Diabetes clinic chains and digital-therapeutics programmes already running connected-CGM coaching. Value: triage of coach and educator time. Willingness to pay is untested.

What a buyer would require

Validation on Indian patients and diets, a prospective pilot with outcome measures, integration with sensor clouds and the clinic EHR (ABDM FHIR).

Regulation and privacy

Likely software as a medical device under India's Medical Devices Rules 2017 (CDSCO draft guidance, Oct 2025). Real data falls under the DPDP Act 2023 and Rules 2025.

Next steps

- Merge the three alert types into one stream with a shared cooldown; set budgets with clinicians
- Noise-augmented training and a noise detector that widens intervals or abstains
- A prospective, observation-only pilot with an Indian diabetes clinic (forecasts hidden from care until validated)
- Interoperability: CGM vendor APIs and ABDM-compliant FHIR exchange

Thank you

Everything shown is in the public repository: code, trained models, the locked test results and the scripts that regenerate every number.

github.com/Spiritsfuse/pratirup

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